Outlier Analysis

The outlier analysis was used to determine the number of times actual returns exceeded a predicted three standard deviation movement. That is:

$$\text{Outlier if } \left|\frac{y_t}{\hat{\sigma}_t}\right| > 3$$

The outlier analysis consists of determining the total number of outliers observed based on the predicted volatility from each model. We chose three standard deviations as the criteria for outliers. If the absolute value of price return for the index was greater than 3 times the forecasted standard deviation for the index on that day the observation was counted as an outlier. The goal of the outlier analysis was to determine which model resulted in the fewest number of surprises.

Results

The HMA-VIX model was found to be a significant improvement over the historical moving average model. It also performed better than the GARCH and EWMA models under various test statistics, and as well as the GARCH and EWMA models in the other tests.

RMSE performance criteria: The HMA-VIX volatility model was the best performing model. The EWMA was the second best model for the SP500 index followed by the GARCH. The GARCH was the second best model for the R2000 index followed by the EWMA. So while the HMA-VIX was the best there was no clear second best. As expected, the simple HMA standard deviation model was the worst performing model for both indexes.

RMZSE performance criteria: The GARCH model was the best performing model for both the SP500 and R2000 indexes. The HMA-VIX was the second best performing model for the R2000 index followed by the EWMA. The EWMA was the second best performing model for the SP500 index followed closely by the HMA-VIX. Therefore, while the GARCH was the best there was no clear second best. The HMA-VIX model performed as well as the EWMA based on the RMZSE analysis.

These testing metrics for the RMSE and RMZSE are shown in Table 6.3.

Outlier performance criteria: The HMA-VIX and GARCH models had the fewest number of outliers (surprises) for predicted SP500 index returns. There were two surprises with these models. The EWMA approach resulted in three outliers and the HMA resulted in five. The GARCH and EWMA models resulted in two outliers each for the R2000 index. The HMA-VIX had three outliers and the HMA had six. Overall, the GARCH model had